

Amit Kumar Maurya

Gorakhpur, U.P. | akmaurya2657@gmail.com | +91 8881345095
[linkedin.com/in/akmaurya2657](https://www.linkedin.com/in/akmaurya2657) | github.com/amitmaurya19

Education

B.Tech (CSE)	Deen Dayal Upadhyaya Gorakhpur University	CGPA: 7.76	2022 – 2026
XII (PCM)	Divine Public School (CBSE)	Percentage: 79.80%	2021
X	Divine Public School (CBSE)	Percentage: 90.67%	2019

Professional Experience

Intern – System and Software Security <i>NIELIT Gorakhpur (ISEA Project Phase-III, Government of India – MeitY)</i> - Selected through competitive interview for a 2-month full-time offline internship focused on national cybersecurity initiatives under the Information Security Education and Awareness (ISEA) framework. - Contributing to system hardening protocols, software vulnerability assessment, and implementing security best practices aligned with government cybersecurity standards. - Conducting research on emerging software vulnerabilities, threat modeling, and developing security protocols for critical system architectures.	Jan 2026 – March 2026
Python Development Intern <i>Vault of Codes</i> - Completed virtual internship centered on Python development, real-world problem-solving, and project-based learning. - Developed multiple hands-on projects including web scraping tools, automation scripts, and API integrations using Python frameworks.	Sept 2024 – Oct 2024

Technical Projects

Off-Grid Peer-to-Peer LoRa Communication System - Engineered mesh-style communication system using two Heltec LoRa 32 V3 boards for long-range (10+ km), internet-independent data transfer. - Developed custom firmware supporting Morse code signaling and integrated external keypad for low-power manual data entry. - Designed secure, SIM-less communication protocols with encryption, suitable for disaster response and off-grid IoT deployments.
Advanced Network Monitoring & Red Teaming Simulation Traffic Analytics: Developed Android application to monitor real-time background network activities with user-defined threshold alerts for data exfiltration detection. Decoy Data Exfiltration: Built specialized app disguised as a game that simulates stealthy background data exfiltration to remote server, testing monitoring system efficacy with obfuscation techniques. HID Attacks: Executed Human Interface Device attacks using programmable USB devices demonstrating keystroke injection, credential harvesting, and privilege escalation vulnerabilities.
Network Intrusion Detection System (NIDS) - Built Python-based NIDS using Scapy for real-time packet capture with deep packet inspection and signature-based detection for common attack vectors. - Designed rule-based anomaly detection flagging port scans, SYN floods, spoofed packets, ARP spoofing, and abnormal traffic patterns. - Implemented comprehensive logging with timestamps and automated PCAP export for forensic review, plus real-time statistics dashboard with protocol breakdown.
Real-Time Chat Platform (ChatVerse) - Built full-featured real-time messaging platform using React.js, Node.js, Express.js, Socket.io, and MongoDB supporting unlimited concurrent chat rooms. - Engineered scalable WebSocket infrastructure managing 100+ concurrent users with auto-join logic, typing indicators, and real-time synchronization. - Deployed on Vercel with JWT authentication, rate limiting, and responsive Tailwind CSS design for seamless cross-device experience.

Intelligent Parking Management System

- Developed IoT-based automated parking solution using ESP32, ultrasonic sensors, and OpenCV for real-time slot detection and license plate recognition.
- Created Flask web dashboard with Firebase integration for cloud-based monitoring, occupancy statistics, and historical usage patterns.

Automated Attendance System with Face Recognition

- Built contactless attendance system using deep learning-based face recognition with OpenCV and face_recognition library, featuring anti-spoofing measures.
- Developed Tkinter GUI with admin panel for managing student records, viewing statistics, and generating automated reports.

Skills

- **Cybersecurity:** Penetration Testing (Metasploit, Nmap, Burp Suite, Wireshark), HID Attacks, Packet Crafting (Scapy), Web Application Security.
- **Languages & Frameworks:** Python (NumPy, Pandas, Flask, Django, TensorFlow), C, C++, JavaScript (React.js, Next.js, Node.js, Express.js), SQL/NoSQL (MongoDB, MySQL).
- **Tools & Platforms:** Linux (Kali, Ubuntu), Git/GitHub, Socket.io, Firebase, Vercel, Android SDK.
- **Leadership:** Co-led Samsung Innovation Campus 2025, managed logistics and data for 1100+ students.

Achievements & Certifications

- **Samsung Innovation Campus 2024 National Topper in Coding & Programming:** Secured All India Rank #1. Awarded 1,00,000 cash prize, Galaxy Book4 Pro 360 laptop, Trophy of Excellence, and featured in Samsung Newsroom India. Visited Samsung India's Head Office and R&D Centers.
- **Geo-Data Sharing and Cyber Security Course (IIRS-ISRO, Dec 2024):** Completed 10-hour certified program focused on secure geo-spatial data handling, satellite data security protocols, and cyber risk fundamentals.
- **Samsung Innovation Campus 2025 IoT Training Completed:** Completed practical IoT training covering sensor interfacing, edge-device programming, and IoT system workflows.
- **Google Cloud Arcade Premium Milestone Achiever:** Achieved Premium level by completing advanced Google Cloud labs on VMs, compute, and cloud operations.
- **Hacktoberfest 2024 Open Source Contributor:** Contributed to global open-source projects through pull requests, maintainer collaboration, and active community participation.